

What's wrong with WebSocket APIs? Unveiling vulnerabilities in WebSocket APIs. (English translation)

What's wrong with WebSocket APIs? Unveiling vulnerabilities in WebSocket APIs. -

@speakerdeck, Mikhail Egorov, Speaker Deck.

- Published: 2019-10-25
- Original: <<https://speakerdeck.com/0ang3el/whats-wrong-with-websocket-apis-unveiling-vulnerabilities-in-websocket-apis>>
- Preserved from: <https://speakerdeck.com/0ang3el/whats-wrong-with-websocket-apis-unveiling-vulnerabilities-in-websocket-apis> (live) on 2026-08-09
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Mikhail Egorov

October 25, 2019

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Transcript

What's wrong with w/ WebSocket API? Unveiling vulnerabilities in WebSocket

APIs Mikhail Egorov / @0ang3el #HACKTIVITY2019

whoami 2 Security researcher / full-time bug hunter

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4 WebSocket protocol essentials

WebSocket protocol – RFC 6455 5 Efficient two-way communication

protocol WebSocket is stateful (HTTP is stateless) Two main parts: handshake and data transfer

WebSocket protocol – RFC 6455 6 Extensibility: subprotocols and

extensions Subprotocols

<https://www.iana.org/assignments/websocket/websocket.xml#subprotocol-name> Wamp
Stomp Soap ...

WebSocket protocol – RFC 6455 7 Extensibility: subprotocols and

extensions Extensions <https://www.iana.org/assignments/websocket/websocket.xml#extension-name> permessage-deflate bbf-usb-protocol

WebSocket protocol – RFC 6455 8 Origin-based security model

(Browser clients) No authentication Client must do client-to-server masking

WebSocket protocol support 9 Major web browsers Web

servers / Proxies Apache httpd, Nginx, IIS, ... HAProxy, Traefik, Varnish, Envoy, ... Cloud
providers WebSocket API (api gateways) WebSocket proxying (load balancers)

WebSocket handshake 10 Upgrade request Base64(Random nonce) Protocol version Required

HTTP version

WebSocket handshake 11 Required status code BASE64(SHA1(Sec-WebSocket-Key || CONST))

WebSocket data transfer 12 \x00 – continuation frame \x01 –

text frame \x02 – binary frame \x08 – close frame \x09 – ping \x0A – pong other values are reserved

WebSocket data transfer - masking 13 Masking key is

32-bit long passed inside frame Client must send masked data $MASKED = MASK \oplus DATA$ ($\oplus$ - XOR) Mechanism protects against cache poisoning and smuggling attacks

14 Cross-Site WebSocket Hijacking

-

WebSocket security for Web Browser 15 SOP doesn't work

for WebSocket in web browser Read from WebSocket cross-origin Write to WebSocket cross-origin Header Origin should be checked on handshake step (origin-based security model)

-

CSWSH 16 Cookies are used to authenticate upgrade request

Header Origin isn't checked or checked poorly

-

CSWSH 17 CORS tricks from @albinowax are applicable to

WebSocket <https://portswigger.net/research/exploiting-cors-misconfigurations-for-bitcoins-and-bounties> Null origin Pre-domain wildcard Post-domain wildcard ...

-

CSWSH – Null origin 18 `nullorigin.html <iframe src="data:text/html, <script>const`

`socket = new WebSocket('wss://example.com'); </script>"></iframe>`

-

CSWSH 19 Playground <https://portswigger.net/web-security/websockets/cross-site-websocket-hijacking>

-

CSWSH – template for attack 5

-

Demo 5

-

22 Authentication / IDOR issues

-

Authentication 23 WebSocket protocol doesn't offer authentication Developers

have to roll out their own AuthN It's secure to check AuthN only during handshake Common secure implementations Session cookies Tokens

-

Broken authentication – Case 1 24 Some ID /

GUID is required in Upgrade request Guess ID Leak GUID (minor IDOR, ...)

-

Broken authentication – Case 2 25 No authentication during

handshake step Some ID / GUID required in API messages Guess ID Leak GUID (minor IDOR, ...)

-

Broken authentication – Case 2 26 Exposing GraphQL subscriptions

w/o AuthN <https://github.com/righettod/poc-graphql#subscriptions-websocket-endpoint-default-enabling> Path /subscriptions

-

Insecure Direct Object Reference issues 27 Strong authentication during

handshake step Some ID / GUID required in API messages Guess ID Leak GUID (minor IDOR, ...)

-

28 Smuggling through WebSocket

-

Reverse proxying WebSocket connection 29 Client Frontend Reverse proxy Backend

/socket.io/ Public WebSocket API

-

Reverse proxying WebSocket connection 30 Client Frontend Reverse proxy Upgrade

request Upgrade request Backend /socket.io/

-

Reverse proxying WebSocket connection 31 Client Frontend Reverse proxy Upgrade

request Upgrade request HTTP/1.1 101 HTTP/1.1 101 Backend /socket.io/

-

Reverse proxying WebSocket connection 32 Client Frontend Reverse proxy Upgrade

request Upgrade request HTTP/1.1 101 HTTP/1.1 101 WebSocket connection direct WebSocket connection Client - Backend Backend /socket.io/

-

Smuggling through WebSocket connection 33 Client Frontend Reverse proxy (vulnerable)

Private REST API Public WebSocket API Backend /internal /socket.io/

-

34 Backend Client Frontend Reverse proxy (vulnerable) /internal Upgrade request

/socket.io/ Sec-WebSocket-Version: 1337 Upgrade request Sec-WebSocket-Version: 1337 Version correctness isn't checked! Smuggling through WebSocket connection

-

35 Backend Client Frontend Reverse proxy (vulnerable) /internal Upgrade request

/socket.io/ Sec-WebSocket-Version: 1337 Upgrade request Sec-WebSocket-Version: 1337 HTTP/1.1 426 HTTP/1.1 426 Response correctness isn't checked! Smuggling through WebSocket connection

-

36 Backend Client Frontend Reverse proxy (vulnerable) /internal Upgrade request

/socket.io/ Sec-WebSocket-Version: 1337 Upgrade request Sec-WebSocket-Version: 1337 HTTP/1.1 426 HTTP/1.1 426 TLS connection direct TLS connection Client – Backend not WebSocket!!! Client can access /internal Smuggling through WebSocket connection

-

Challenge – challenge.0ang3el.tk 37 URL <https://challenge.0ang3el.tk/websocket.html> You

need to access flag on localhost:5000 Seems no one solved

-

Challenge – challenge.0ang3el.tk 38 Frontend Not disclosed WebSocket

reverse proxy socket.io.js Proxies only WebSocket API - /socket.io/ path Backend Flask, Flask-SocketIO, Flask-Restful Listens on localhost:5000 only

-

challenge1.py

-

challenge1.py - DEMO

-

Vulnerable reverse proxies 41 Vulnerable Varnish, Envoy proxy

<= 1.8.0, other non-disclosed Not vulnerable Nginx, HAProxy, Traefik, others

-

Varnish response 42 WebSocket proxying configuration <https://varnish-cache.org/docs/6.3/users-guide/vcl-example-websockets.html>

-

Smuggling through WebSocket connection 43 Client Frontend Reverse proxy (Nginx

or another) Private REST API Public WebSocket API & REST API Backend /internal /api/socket.io/ /api/health

-

Smuggling through WebSocket connection 44 Client Frontend Reverse proxy (Nginx

or another) Backend /internal /api/socket.io/ /api/health example.com GET HTTP/1.1 200

-

Smuggling through WebSocket connection 45 Client Frontend Reverse proxy (Nginx

or another) Backend /internal /api/socket.io/ /api/health Only Upgrade: websocket header is checked! POST /api/health?u= POST /api/health?u=

-

Smuggling through WebSocket connection 46 Client Frontend Reverse proxy (Nginx

or another) Backend /internal /api/socket.io/ /api/health attacker.com GET HTTP/1.1 101 HTTP/1.1 101 HTTP/1.1 101 Only status code is checked for response! POST /api/health?u= POST /api/health?u=

Smuggling through WebSocket connection 47 Client Frontend Reverse proxy (Nginx

or another) Backend /internal /api/socket.io/ /api/health HTTP/1.1 101 HTTP/1.1 101 TLS connection direct TLS connection Client – Backend not WebSocket!!! Client can access /internal POST /api/health?u= POST /api/health?u= Client-to-Server masking isn't checked by proxy!!!

Challenge2 – challenge2.0ang3el.tk 48 URL <https://challenge2.0ang3el.tk/websocket.html> You

need to access flag on localhost:5000 Seems no one solved

Challenge2 – challenge2.0ang3el.tk 49 Frontend Nginx as WebSocket

reverse proxy socket.io.js Proxies only /api/public path (socket.io and healthcheck) Backend Flask, Flask-SocketIO, Flask-Restful Listens on localhost:5000 only

Challenge2 – challenge2.0ang3el.tk 50 Nginx config

Challenge2 – challenge2.0ang3el.tk 51 REST API - healthcheck

Challenge2.py 5

Challenge2.py - Demo 5

Vulnerable reverse proxies 54 Almost all proxies are affected

But exploitation is limited External SSRF is required that returns status code ...

55 Discovering WebSocket APIs

-

Discovering WebSocket API 56 Monitor Upgrade requests Analyze

JavaScript files Try to establish WebSocket connection to each URL ...

-

57 Conclusion

-

Ideas for further research 58 Security of WebSocket subprotocols

More smuggling techniques HTTP/2 and WebSocket ...

-

Thank you! @0ang3el